

```
>> dynare osr.mod
Starting Dynare (version 6.4).
Calling Dynare with arguments: none
Starting preprocessing of the model file ...
Found 23 equation(s).
Evaluating expressions...
Computing static model derivatives (order 1).
Normalizing the static model...
Finding the optimal block decomposition of the static model...
7 block(s) found:
    5 recursive block(s) and 2 simultaneous block(s).
    the largest simultaneous block has 16 equation(s)
                                   and 16 feedback variable(s).
Computing dynamic model derivatives (order 1).
Normalizing the dynamic model...
Finding the optimal block decomposition of the dynamic model...
5 block(s) found:
    3 recursive block(s) and 2 simultaneous block(s).
    the largest simultaneous block has 16 equation(s)
                                   and 16 feedback variable(s).

Preprocessing completed.
Preprocessing time: 0h00m00s.
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STEADY-STATE RESULTS:

q	29.8013
k	1.46248
R	1.01516
Rp	1.0121
b	42.9331
x	0.0137035
varphi	0.0317528
mu	0.00522869
kp	2.67752
chi	1.02332
nu	0.00416305
eta	1.38229
zeta	1.02332
N	11.7043
phi	0.272618
Ne	11.6419
Nn	0.0624575
xp	2.27908
bp	31.2288
Y	2.29278
eb	0
eN	0
eY	0

EIGENVALUES:

Modulus	Real	Imaginary
1.243e-19	-1.243e-19	0
7.79e-18	-7.79e-18	0
2.289e-17	-2.289e-17	0
1.217e-16	-1.217e-16	0
8.606e-16	8.606e-16	0
1.933e-15	-1.933e-15	0
0.1789	-0.1789	0

0.1934	0.1934	0
0.8	0.8	0
0.8	0.8	0
0.9822	0.9822	0
1.019	1.019	0
1.02	1.02	0
1.021	1.021	0
1.023	1.023	0
5.261e+33	-5.261e+33	0
Inf	-Inf	0
Inf	-Inf	0

There are 7 eigenvalue(s) larger than 1 in modulus for 7 forward-looking variable(s)
The rank condition is verified.

MODEL SUMMARY

Number of variables: 23
Number of stochastic shocks: 2
Number of state variables: 11
Number of jumpers: 7
Number of static variables: 7

MATRIX OF COVARIANCE OF EXOGENOUS SHOCKS

Variables	ep_N	ep_Y
ep_N	0.000027	0.000000
ep_Y	0.000000	0.000709

POLICY AND TRANSITION FUNCTIONS

	Y	q	b	N	
bp		phi	eb	eN	
Constant		29.801321	42.933083	11.704322	31.228762
2.292782	0.272618	0	0		
k(-1)		0.719571	80.952488	0	80.952488
30.255691	-0.514035	0	0		
R(-1)		-0.313209	-3.955083	41.730957	
-45.686040	0	0.997114	0	0	
Rp(-1)		-0.515121	-80.705076	-30.354356	-50.350720
-0.327721	-0.194552	0	0		
b(-1)		-0.024151	-2.717012	0	-2.717012
-1.015159	0.017253	0	0		
kp(-1)		0	0	0	0
30.102344	0	0	0		
N(-1)		0.012027	0.287754	0.994664	-0.706909
1.023317	0.021341	0.055202	0		
phi(-1)		-0.007943	-1.244399	-0.468037	-0.776363
-0.481519	-0.003000	0	0		
bp(-1)		0.000034	0.005318	0.002000	0.003318
1.012101	0.000013	0	0		
eN(-1)		-0.389029	10.212603	-9.313491	19.526094
-9.313491	-0.281779	0	0.800000		
q(-1)		-0.006283	-3.051242	0	
-3.051242	0	0.019375	0.071467	0	
eY(-1)		-0.962963	-6.645642	0	-6.645642
-1.834226	0.042199	0	0		
ep_N		-0.486286	12.765753	-11.641864	24.407617

-11.641864	-0.352223	0	1.000000		
ep_Y		-1.203704	-8.307053	0	-8.307053
-2.292782	0.052748	0	0		

THEORETICAL MOMENTS

VARIABLE	MEAN	STD. DEV.	VARIANCE
q	29.8013	0.0575	0.0033
b	42.9331	0.3852	0.1484
N	11.7043	1.6189	2.6210
bp	31.2288	1.3924	1.9388
Y	2.2928	0.1434	0.0206
phi	0.2726	0.0361	0.0013
eb	0.0000	0.0912	0.0083
eN	0.0000	0.0086	0.0001

VARIANCE DECOMPOSITION (in percent)

	ep_N	ep_Y
q	11.05	88.95
b	51.03	48.97
N	92.10	7.90
bp	90.37	9.63
Y	48.84	51.16
phi	92.23	7.77
eb	91.26	8.74
eN	100.00	0.00

MATRIX OF CORRELATIONS

Variables	q	b	N	bp	Y	phi	eb	eN
q	1.0000	0.7481	0.4264	-0.2888	0.7771	0.3944	0.4347	-0.1466
b	0.7481	1.0000	0.6659	-0.4976	0.3194	0.6274	0.6853	0.1524
N	0.4264	0.6659	1.0000	-0.9785	0.2818	0.9987	0.9979	-0.2944
bp	-0.2888	-0.4976	-0.9785	1.0000	-0.2393	-0.9876	-0.9706	0.3845
Y	0.7771	0.3194	0.2818	-0.2393	1.0000	0.2725	0.2506	-0.6988
phi	0.3944	0.6274	0.9987	-0.9876	0.2725	1.0000	0.9952	-0.3176
eb	0.4347	0.6853	0.9979	-0.9706	0.2506	0.9952	1.0000	-0.2361
eN	-0.1466	0.1524	-0.2944	0.3845	-0.6988	-0.3176	-0.2361	1.0000

OPTIMAL SIMPLE RULE

OSR: Initial value of the objective function: 0.0205706

f at the beginning of new iteration, 0.0205706264

Predicted improvement: 0.000000000

lambda =	1;	f =	0.0205706
lambda =	1.9332;	f =	0.0205706
lambda =	3.7372;	f =	0.0205706
lambda =	7.2247;	f =	0.0205706
lambda =	13.967;	f =	0.0205706
lambda =	27;	f =	0.0205706
lambda =	52.196;	f =	0.0205706
lambda =	100.9;	f =	0.0205706
lambda =	195.07;	f =	0.0205706

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lambda =      377.1; f =      0.0205706
lambda =      729; f =      0.0205705
lambda =    1409.3; f =      0.0205705
lambda =    2724.4; f =      0.0205703
lambda =    5266.8; f =      0.0205700
lambda =    10182; f =      0.0205695
lambda =    19683; f =      0.0205684
lambda =    38051; f =      0.0205663
lambda =    73559; f =      0.0205622
lambda =  1.422e+05; f =      0.0205545
lambda =  2.749e+05; f =      0.0205398
lambda =  5.3144e+05; f =      0.0205124
lambda =  1.0274e+06; f =      0.0204627
lambda =  1.9861e+06; f =      0.0203777
lambda =  3.8395e+06; f =      0.0202480
lambda =  7.4224e+06; f =      0.0200894
Norm of dx 1.0709e-07
----
Improvement on iteration 1 =      0.000481191
-----
f at the beginning of new iteration,      0.0200894350
Predicted improvement:      0.000044397
lambda =      1; f =      0.0200213
lambda =    1.9332; f =      0.0199871
Norm of dx    0.30151
----
Improvement on iteration 2 =      0.000102314
-----
f at the beginning of new iteration,      0.0199871206
Predicted improvement:      0.000009056
lambda =      1; f =      0.0199749
Norm of dx    0.22257
----
Improvement on iteration 3 =      0.000012245
-----
f at the beginning of new iteration,      0.0199748758
Predicted improvement:      0.000002049
lambda =      1; f =      0.0199725
Norm of dx    0.134
----
Improvement on iteration 4 =      0.000002349
-----
f at the beginning of new iteration,      0.0199725264
Predicted improvement:      0.000000066
lambda =      1; f =      0.0199725
Norm of dx    0.026377
----
Improvement on iteration 5 =      0.000000070
improvement < crit termination

OPTIMAL VALUE OF THE PARAMETERS
Variables      Value
gamma_q      2.225697
gamma_N      2.404180

Objective function :      0.0199725

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MODEL SUMMARY

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MATRIX OF COVARIANCE OF EXOGENOUS SHOCKS

Variables	ep_N	ep_Y
ep_N	0.000027	0.000000
ep_Y	0.000000	0.000709

POLICY AND TRANSITION FUNCTIONS

bp	Y	q	b	N	eY
Constant		29.801321	42.933083	11.704322	31.228762
2.292782	0.272618	0	0	0	0
k(-1)		-5.194023	65.217390	0	65.217390
30.255691	-0.414119	0	0	0	0
R(-1)		-0.238743	-3.734656	41.730957	
-45.465613	0	0.995714	0	0	0
Rp(-1)		5.536389	-64.619215	-30.354356	-34.264859
-0.327721	-0.296694	0	0	0	0
b(-1)		0.174327	-2.188894	0	-2.188894
-1.015159	0.013899	0	0	0	0
kp(-1)		0	0	0	0
30.102344	0	0	0	0	0
N(-1)		-0.191151	-6.630202	0.994664	-7.624866
1.023317	0.065268	0.205410	0	0	0
phi(-1)		0.085366	-0.996370	-0.468037	-0.528333
-0.481519	-0.004575	0	0	0	0
bp(-1)		-0.000365	0.004258	0.002000	0.002258
1.012101	0.000020	0	0	0	0
eN(-1)		8.094081	32.839527	-9.313491	42.153018
-9.313491	-0.425456	0	0.800000	0	0
q(-1)		-0.003539	-3.180550	0	0
-3.180550	0	0.020196	0.074685	0	0
eY(-1)		-0.476509	-5.349296	0	-5.349296
-1.834226	0.033967	0	0	0.800000	0
ep_N		10.117601	41.049408	-11.641864	52.691272
-11.641864	-0.531820	0	1.000000	0	0
ep_Y		-0.595636	-6.686621	0	-6.686621
-2.292782	0.042459	0	0	1.000000	0

THEORETICAL MOMENTS

VARIABLE	MEAN	STD. DEV.	VARIANCE
q	29.8013	0.2675	0.0716
b	42.9331	5.0621	25.6254
N	11.7043	0.9597	0.9211
bp	31.2288	6.0134	36.1609
Y	2.2928	0.1413	0.0200
phi	0.2726	0.0544	0.0030
eb	0.0000	0.1783	0.0318
eN	0.0000	0.0086	0.0001

eY	0.0000	0.0444	0.0020
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VARIANCE DECOMPOSITION (in percent)

	ep_N	ep_Y
q	95.29	4.71
b	92.10	7.90
N	92.18	7.82
bp	92.16	7.84
Y	48.71	51.29
phi	92.22	7.78
eb	91.70	8.30
eN	100.00	0.00
eY	-0.00	100.00

MATRIX OF CORRELATIONS

Variables	q	b	N	bp	Y	phi	eb	eN	eY
q	1.0000	0.9248	-0.9467	0.9296	-0.2393	-0.9362	-0.9152	0.5792	-0.0243
b	0.9248	1.0000	-0.9895	0.9997	-0.0397	-0.9982	-0.9968	0.2547	0.0425
N	-0.9467	-0.9895	1.0000	-0.9926	0.1323	0.9963	0.9931	-0.3158	-0.1070
bp	0.9296	0.9997	-0.9926	1.0000	-0.0545	-0.9993	-0.9977	0.2649	0.0529
Y	-0.2393	-0.0397	0.1323	-0.0545	1.0000	0.0779	0.0597	-0.6785	-0.7145
phi	-0.9362	-0.9982	0.9963	-0.9993	0.0779	1.0000	0.9979	-0.2805	-0.0692
eb	-0.9152	-0.9968	0.9931	-0.9977	0.0597	0.9979	1.0000	-0.2274	-0.0968
eN	0.5792	0.2547	-0.3158	0.2649	-0.6785	-0.2805	-0.2274	1.0000	0.0000
eY	-0.0243	0.0425	-0.1070	0.0529	-0.7145	-0.0692	-0.0968	0.0000	1.0000

Total computing time : 0h00m04s

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